Datathon 2026
Cat B - Afternoon DSA
Enhancing Survey Insight and Effectiveness

1. Introduction

Surveys remain one of the most widely used tools for organisations to gather insights on attitudes, preferences and behaviours. However, despite their ubiquity, many surveys suffer from persistent issues such as low completion rates, redundant or poorly designed questions and difficulty translating raw responses into actionable insights. As survey length and complexity increase, respondent fatigue becomes a critical concern, often resulting in partial or incomplete responses that reduce data quality and bias findings.

In response to these challenges, this project leveraged data visualisation, statistical techniques, machine learning, and supervised learning methods to analyse survey responses, identify weaknesses in survey design and synthesise meaningful insights. Ultimately, it seeks to bridge the gap between raw survey data and practical decision-making.

2. Dataset Overview

The dataset used in this project consists of individual-level survey responses collected from applicants evaluating employer attractiveness. It contains a mixture of demographic variables, behavioural indicators, Likert-scale ratings and survey completion metadata. Key components of the dataset include:

- Demographic attributes, such as age group, education level, major and gender.
- Employer attractiveness ratings, capturing respondents' perceptions of the organization across multiple dimensions.
- Completion status, indicating whether a respondent completed or partially exited the survey.

The dataset reflects a realistic operational survey environment, where responses are heterogeneous, missing data is common and high-dimensional question formats, such as "pick-3" combinatorial choice questions, pose challenges for interpretation and analysis. As such, it provides a suitable foundation for testing analytical approaches aimed at improving survey quality and insight extraction.

3. Methodology

The analytical workflow was designed to align closely with the problem objectives, progressing from survey diagnostics to respondent understanding and insight synthesis.

3.1 Data Cleaning and Preprocessing

Initial preprocessing focused on removing personally identifiable information and irrelevant administrative fields to ensure data privacy and analytical clarity. Categorical variables were standardised, missing values were handled appropriately depending on context, and numerical features were scaled where required. Completion status was treated as a key outcome variable for downstream analysis.

3.2 Survey Quality Diagnostics

To identify opportunities for survey improvement, question-level performance was evaluated using:

- Missingness and non-response rates, highlighting questions that respondents frequently skip.

- Redundancy analysis, using measures of association (e.g. Cramér's V) to detect highly correlated questions that may be duplicative or conceptually overlapping.

These diagnostics allow survey designers to pinpoint questions that contribute disproportionately to respondent burden without adding commensurate informational value, and provide an analytical basis for streamlining survey questionnaires and associated processes in future iterations.

3.3 Completion Behaviour Analysis

To understand factors associated with partial or incomplete survey responses, the project examines relationships between respondent characteristics, survey attributes, and completion outcomes. Interpretable statistical and machine learning models are used to identify variables most strongly associated with survey dropout or completion. Emphasis is placed on explainability, ensuring insights can be communicated clearly to non-technical stakeholders.

3.4 Respondent Segmentation

Following an initial review of partial and incomplete survey responses, the dataset was filtered to retain only completed responses to ensure the robustness and interpretability of subsequent analyses. Supervised learning and data visualisation techniques were then applied to identify meaningful demographic and behavioural segments among respondents.

The core analysis focused on identifying factors influencing the organisation's attractiveness rating by examining correlations between attractiveness scores and variables such as respondents' motivations for working with the organisation, perceptions of the organisation as an employer, and the top three factors respondents expressed interest in learning more about.

3.5 Insight Synthesis

Finally, the results were thoroughly analysed and translated into structured insights and recommendations. Such information allows the company to inform and strengthen the organisation's recruitment and branding strategies to suit potential applicants' demands.

4. Results

4.1 Missingness rate

	question	**missing_rate**
0	Other - Write In (Required):What do you wish to...	0.998087
1	Other - Write In (Required):What do you wish to...	0.998087
2	Other - Write In (Required):Which of these fac...	0.997322
3	Application and interview process:What do you ...	0.795715
4	Work-life balance and culture:What do you wish...	0.611706
5	Compensation and benefits:What do you wish to ...	0.544759
6	Career progression and development:What do you...	0.469396
7	Types of roles available:What do you wish to l...	0.459449
8	On a scale from 1 to 10 (1 – Low, 10 – High), how would you rate the attractiveness of (Employer) as an employer?	0.293037
9	Which of these factors would most motivate you...	0.293037
10	Which of these statements best describes your perception...	0.292655
11	What is your gender?	0.092196

Table 1: Top 12 Questions with the Highest Missingness Rates

From Table 1, we observe the 12 survey questions with the highest missingness rates, with key terms for each question underlined. As expected, questions that are optional or conditionally displayed (e.g. only shown when "Other" is selected) exhibit extremely high missingness rates of approximately 99%, as most respondents are not required to answer them. This is followed by options from "pick-3" question formats, where respondents are required to select only three out of five options, naturally resulting in higher levels of missing data for the remaining choices.

Consequently, the highest-ranking questions do not yield meaningful analytical insights and are excluded from further consideration. The subsequent analysis therefore focuses on the remaining questions within the top 12 that exhibit high missingness but may provide more substantive insights into respondent behaviour and survey design.

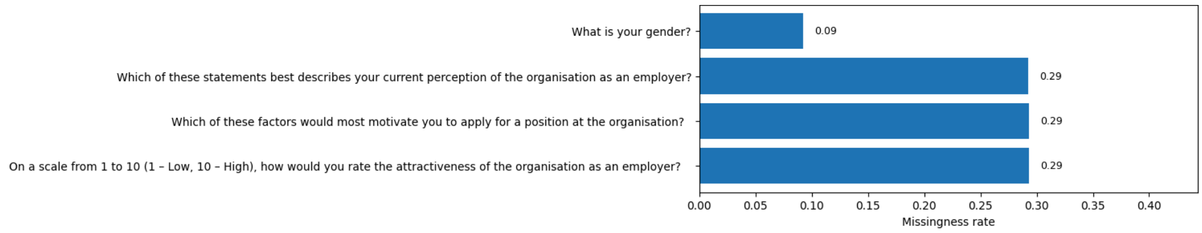

Figure 1: Top 4 Questions with the Highest Missingness Rates after Excluding Non-Informative Items

From Figure 1, we observe questions related to employer attractiveness, motivating factors, and perceptions of the organisation as an employer exhibit similarly high missingness rates of approximately 29%. Given that these questions are central to the survey's objectives, the elevated missingness suggests potential respondent fatigue or cognitive burden at this stage of the questionnaire. These questions likely require greater reflection and subjective evaluation compared to factual items, which may lead some respondents to skip them, especially since they appear later in the survey flow. The similarity in missingness across these three conceptually related questions also indicates a consistent pattern rather than an isolated design flaw, suggesting that question placement, wording complexity, or perceived redundancy may be contributing factors.

In contrast, gender (Question 11) shows a substantially lower missingness rate (approximately 9%), indicating higher respondent willingness to provide demographic information. This aligns with expectations, as demographic questions are typically perceived as straightforward and require minimal cognitive effort. The relatively lower missingness here reinforces the interpretation that missingness in the preceding questions is driven more by survey fatigue or evaluative burden rather than general disengagement.

Overall, these findings highlight that while the core analytical questions are valuable for understanding employer attractiveness, they may benefit from improved survey design, such as clearer phrasing, reduced redundancy, or repositioning earlier in the survey. Nevertheless, it is worth noting the missingness rate from these questions remains moderate, even at approximately 29%.

4.2 Redundancy analysis using Cramér's V

	q1	q2	cramers_v
0	Which higher education institution do you or did you study at?	What will be your highest qualification when you graduate?	0.287
1	Which higher education institution do you or did you study at?	Which of the following best describes the main subject that you are studying?	0.262

Table 2:Top Pairs of Survey Questions Exhibiting Moderate Association (Cramér's V)

Table 2 presents pairs of categorical survey questions that exhibit moderate associations, as measured by Cramér's V. Using a threshold of $0.25 \leq V < 0.95$, the analysis highlights question pairs that may capture overlapping respondent information while avoiding trivially identical or near-duplicate items. This approach provides a quantitative basis for identifying potential redundancies in survey design.

The strongest association (Cramér's V = 0.287) is observed between respondents' higher education institution and their highest qualification upon graduation. This relationship is intuitive, as certain institutions are more strongly associated with specific qualification pathways, leading to structured patterns in respondents' answers. Similarly, a moderate association (Cramér's V = 0.262) is identified between higher education institution and primary field of study, reflecting institutional specialisation and programme offerings that influence students' academic trajectories.

Importantly, while these associations are non-trivial, their magnitudes remain well below levels that would indicate near-duplication. This suggests that the questions, although related, are not interchangeable and still capture distinct dimensions of respondents' educational backgrounds. From a survey design perspective, these findings do not necessarily warrant the removal of any single question. However, they highlight opportunities to streamline the questionnaire by reviewing whether all related items are required in contexts where respondent burden is a concern.

4.3 Factors Associated with Survey Completion

	feature	coef	odds_ratio
61	Please indicate your nationality._Singaporean/...	1.872228	6.502771
67	Which of these statements best describes your current perception ...	1.793675	6.011504
68	Which of these statements best describes your current perception ...	1.780450	5.932526
69	Which of these statements best describes your current perception ...	1.565127	4.783282
32	What will be your highest qualification when y...	1.273688	3.574008
70	Which of these statements best describes your current perception ...	1.113801	3.045914
30	What is your current year of study as of 2025?...	0.318532	1.375108

27	What is your current year of study as of 2025?...	0.272572	1.313338
12	Which higher education institution do you or ...	0.270242	1.310282
196	User Agent_Mozilla/5.0 (Windows NT 10.0; Win64...	0.258516	1.295007

Table 3: Variables Positively Associated with Survey Completion (Logistic Regression Results)

The results summarised above present the strongest positive predictors of survey completion, based on a logistic regression model. Coefficients and corresponding odds ratios indicate the direction and magnitude of association between respondent characteristics and the likelihood of completing the survey, holding other factors constant.

Several demographic and perceptual variables exhibit strong positive associations with survey completion. In particular, respondents identified as Singaporean nationals show substantially higher odds of completing the survey (odds ratio ≈ 6.50), suggesting greater engagement or relevance of the survey content to this group. Similarly, multiple items relating to respondents' perceptions of the organisation are among the strongest positive predictors, with odds ratios ranging from approximately 3.0 to 6.0. This pattern indicates that respondents who hold clearer or more favourable views about the organisation are significantly more likely to complete the survey, reinforcing the link between perceived relevance and sustained engagement.

Educational characteristics also play a role. Respondents' expected highest qualification and year of study are positively associated with survey completion, albeit with more moderate effect sizes. This suggests that respondents further along in their educational trajectory may be more accustomed to participating in surveys or more motivated to provide complete responses, particularly as they are more actively considering employment opportunities and career-related decisions.

	feature	coef	odds_ratio
158	User Agent_Mozilla/5.0 (Linux; Android 15; K) ...	-1.034232	0.355499
59	Please indicate your nationality._Non-Singapor...	-0.837497	0.432792
60	Please indicate your nationality._Non-Singapor...	-0.738747	0.477712
18	Which higher education institution do you or...	-0.591486	0.553504
26	What is your current year of study as of 2025?...	-0.527730	0.589943
36	What will be your highest qualification when y...	-0.456911	0.633237
40	Which of the following best describes the main subject...	-0.417407	0.658752
34	What will be your highest qualification when y...	-0.299439	0.741234
33	What will be your highest qualification when y...	-0.166709	0.846446

Table 4: Variables Negatively Associated with Survey Completion (Logistic Regression Results)

On the other hand, the strongest negative associations with survey completion are observed among specific demographic and technical factors. For instance, non-Singaporean nationality is consistently associated with lower completion odds, suggesting that parts of the survey may be less relevant, accessible, or engaging for international respondents.

Technical factors, such as user agent and device type, also appear among the negative predictors. Respondents accessing the survey via certain mobile or non-desktop environments show lower completion probabilities, which may reflect usability constraints, survey length, or interface friction on smaller screens. To mitigate this, future survey iterations could prioritise mobile-first design principles, reduce question density per screen and implement clearer progress indicators to improve completion rates across device types.

	Current year of study	**Main subject**	**n**	**avg_p_complete**	**completion _rate**
1	Year 3	Economics	20	0.413124	0.400000
2	Year 1	IT and Technology	53	0.605064	0.603774
3	Year 1	Mathematical Science/Statistics	145	0.607308	0.606897
4	Year 3	Mathematical Science/Statistics	81	0.643475	0.641975
5	Year 3	Engineering (Electrical/Electronics, Mechanica...	32	0.662887	0.656250
6	Year 2	IT and Technology	54	0.684306	0.685185
7	Year 2	Mathematical Science/Statistics	128	0.718142	0.718750
8	Year 1	Natural Science (Physics, Chemistry, Biology, ...	197	0.735877	0.736041
9	Year 1	Architecture, Building and Planning	23	0.739812	0.739130
10	Year 2	Natural Science (Physics, Chemistry, Biology, ...	155	0.755525	0.754839
11	Year 1	Business/Management (Accounting, Finance, Mark...	91	0.757998	0.758242
12	Year 3	Medicine, Dentistry and related subjects	38	0.764047	0.763158
13	Year 3	IT and Technology	36	0.775969	0.777778
14	Year 4	Mathematical Science/Statistics	43	0.790747	0.790698

Table 5:Academic Profiles with the Lowest Survey Completion Rates

Table 5 summarises survey completion patterns across combinations of respondents' year of study and primary field of study, among the respondents who have completed the questions on their year of study and main subject. The table reports both predicted probabilities of completion (average_p_complete) and observed completion rates (completion_rate). The results reveal clear and systematic differences in completion behaviour across academic profiles, suggesting that educational context plays a meaningful role in sustained survey engagement.

A clear gradient by year of study is observed across multiple disciplines. Respondents in earlier years of study, particularly Year 1, generally exhibit lower completion rates compared to those in Year 2 and above. This trend is most evident within mathematically intensive and technical disciplines, where completion rates increase steadily from Year 1 through Year 4. For example, students in Mathematical Science and Statistics demonstrate progressively higher completion rates as they advance in their academic trajectory, culminating in the highest observed completion among Year 4 students. This pattern is consistent with earlier findings that respondents further along in their studies are more likely to complete the survey, potentially due to greater familiarity with surveys or increased relevance of the survey topic to their career planning.

Differences by field of study are also apparent. Students in quantitatively oriented disciplines, such as Mathematical Sciences, Natural Sciences, and Engineering, tend to exhibit relatively high completion rates across most years of study. In contrast, certain discipline–year combinations, such as Year 3 Economics, show notably lower completion rates, though these estimates should be interpreted with caution due to smaller sample sizes. These variations may reflect differences in perceived relevance of the organisation to respondents' fields of study and alignment between survey content and academic or career interests.

Overall, the findings indicate that survey completion behaviour is not random but is jointly shaped by respondent characteristics and survey structure. Academic seniority and disciplinary context influence completion likelihood, with early-year students and certain fields exhibiting systematically lower completion rates. In addition, longer or repetitive sections are associated with higher dropout, while respondents holding stronger opinions—either positive or negative—are more likely to complete the survey. Together, these results highlight clear opportunities for targeted survey optimisation, such as repositioning core analytical questions earlier in the survey, consolidating or removing overlapping items, improving mobile usability, and introducing tailored framing or incentives for respondent groups with historically lower completion rates to improve overall response completeness and reduce attrition.

4.4 Analysis of Employer Attractiveness Rating

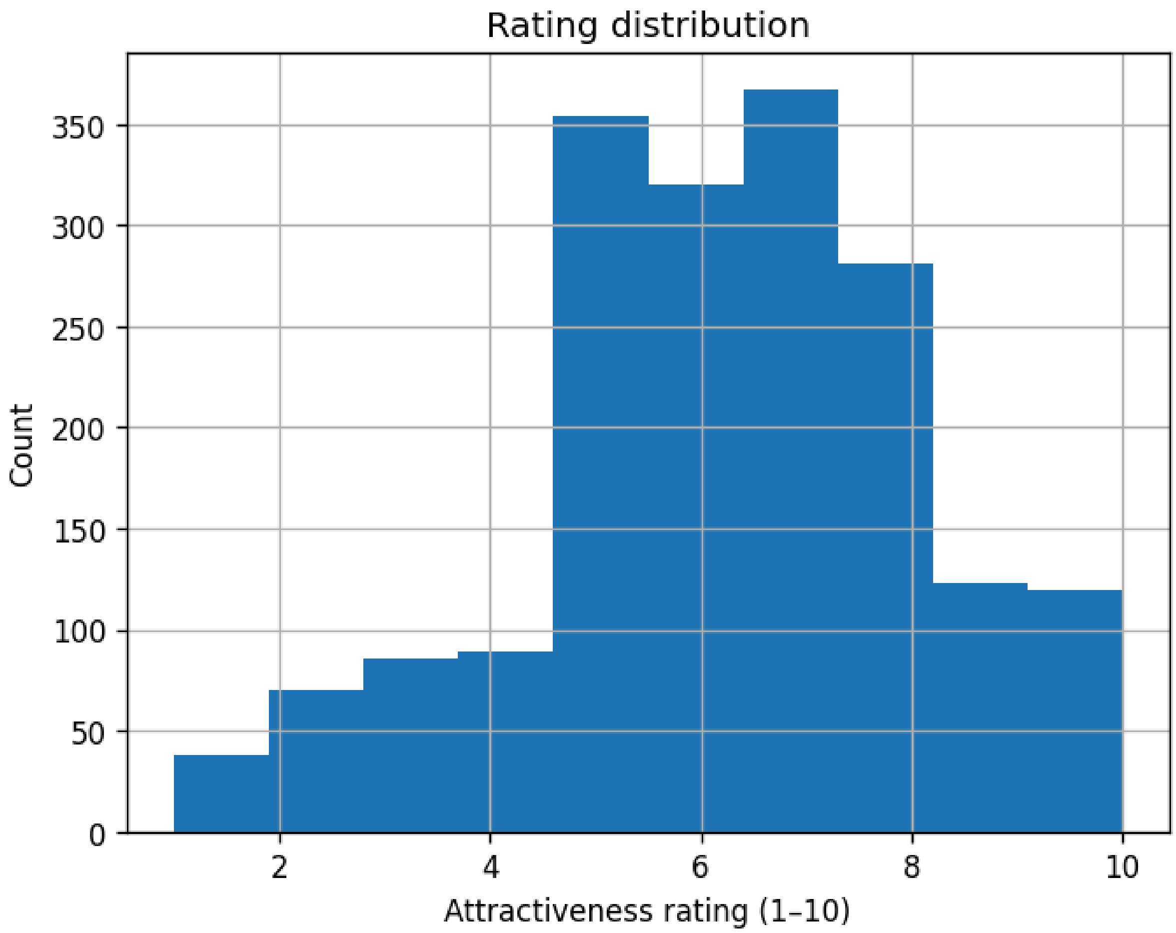

Figure 2:Attractiveness Rating Distribution

The histogram (Figure 2) displays the distribution of employer attractiveness ratings from 1-10 across respondents. The ratings are concentrated in the mid-to-high range, with a clear peak around 5–7, fewer extreme low ratings (1–2) and a right tail extending towards 9–10. This indicates that most respondents perceive the employer as moderately to fairly attractive, with relatively fewer cases of strong dissatisfaction or exceptional appeal.

The dominance of mid-range ratings implies that the employer meets minimum expectations but often fails to clearly differentiate themselves. The relatively small but important high-rating tail (8–10) represents respondents who perceive the employer as genuinely attractive. Comparing their survey responses against the mid-range group could help isolate what factors drive high attractiveness. Conversely, investigating the low-rating tail provides insight into deal-breakers or missing fundamentals that actively repel applicants.

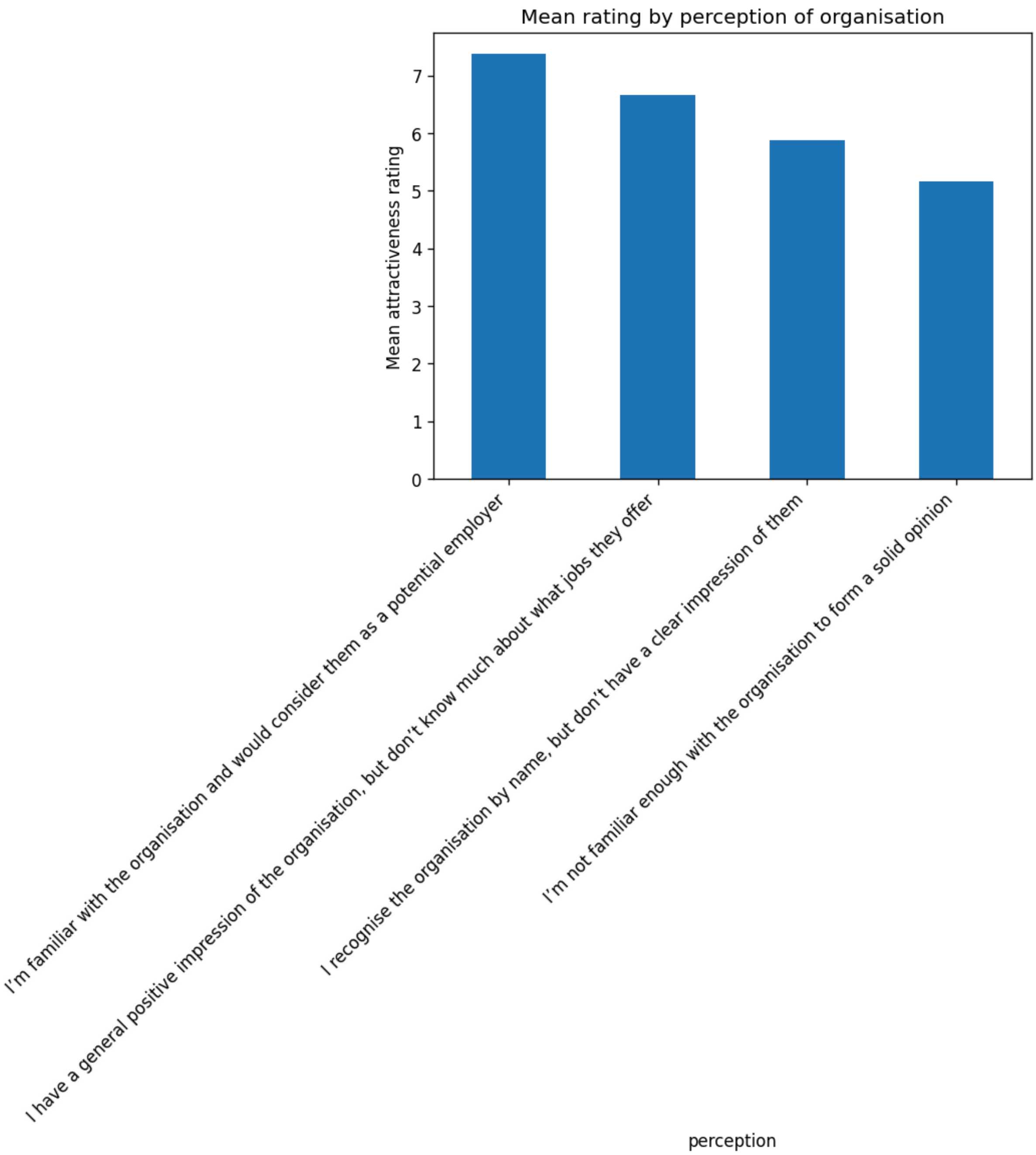

Figure 3: Mean Attractiveness Ratings by Respondents' Perception of the Organisation

Figure 3 plots mean employer attractiveness rating against respondents' current perception of the organisation. The pattern appears strongly monotonic: respondents who are familiar with the organisation and would consider it as a potential employer give the highest ratings; those with a general positive impression but limited job knowledge rate the employer lower; respondents who only recognise the organisation by name rate it lower still; and those not familiar enough to form an opinion give the lowest ratings.

This means that employer attractiveness is driven primarily by familiarity and clarity, not just brand recognition. Increasing organization awareness of respondents appears to yield substantial gains in attractiveness rating. The figure thus suggests building company perception and visibility as a key lever for improving employer attractiveness to potential applicants.

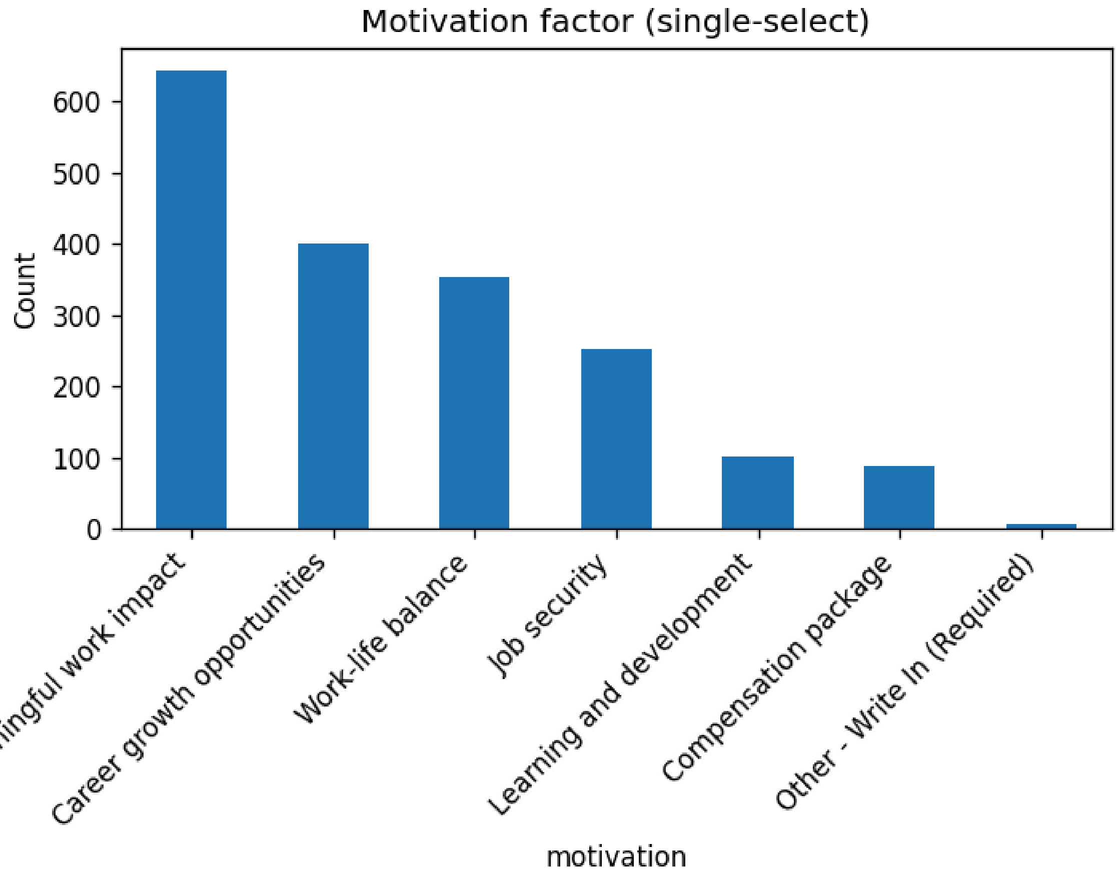

Figure 4: Key Motivation Factors

Figure 4 shows the distribution of the single most important motivation factor respondents consider when deciding whether to apply. From this data we can derive that meaningful work impact is the most dominant driver by a wide margin, exceeding all other factors, though career growth opportunities and work-life balance form a strong second tier. On the other hand, compensation packages alongside learning and development rank substantially lower when respondents are forced to choose only one factor.

This tells us that employer attractiveness in this dataset is not primarily compensation-led. Applicants are more strongly motivated by purpose, long-term career progression and quality of life. This directly informs employer branding, suggesting that ideas such as impact, career growth, and balance should be emphasized in job listings and broader company messaging to improve employer branding.

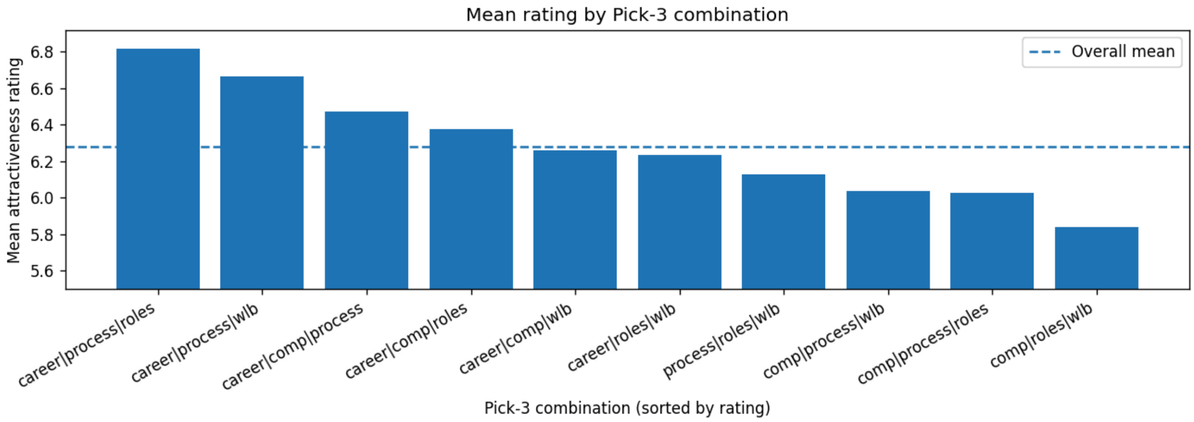

Figure 5: Mean Employer Attractiveness Ratings by Pick-3 Information Preference Combination

Figure 5 presents the ten possible Pick-3 combinations derived from responses to the question *"What do you wish to learn more about regarding the organisation as an employer?"*, covering the options: roles, career progression, compensation, work-life balance and the application and interview process. For each combination, the figure reports the mean employer attractiveness rating among respondents who selected that set of priorities, with combinations ordered by attractiveness and benchmarked against the overall mean.

The results indicate a clear association between respondents' information priorities and their perceptions of employer attractiveness. Combinations that emphasise career development, application processes and role clarity are consistently associated with higher mean attractiveness ratings and lie above the overall average. Notably, the top-ranked combinations all include both career progression and application-related information, suggesting that respondents who value clarity around hiring processes and long-term career trajectories tend to view the organisation more favourably as an employer.

In contrast, combinations that prioritise compensation, particularly when not paired with career or role-related information, are associated with lower attractiveness ratings and fall below the overall mean. This pattern suggests that respondents who are less positively disposed toward the organisation may place greater emphasis on more tangible or immediately comparable factors, such as remuneration, in the absence of stronger signals about growth or role fit.

Overall, these findings highlight that multi-select information preferences serve as a meaningful behavioural signal of employer attractiveness. Reducing informational ambiguity through clearer role descriptions, transparent hiring timelines, explicit interview expectations and well-articulated career pathways may therefore improve employer perceptions by enabling prospective applicants to form more confident and informed evaluations of the organisation.

6. Conclusion

This project presents an analytical process designed to address key challenges in survey analysis and design. By integrating survey diagnostics, completion behaviour analysis, respondent segmentation and automated insight extraction, the solution demonstrates how organisations can move beyond surface-level statistics to better understand respondent behaviour, improve survey quality, and extract actionable insights in a scalable and interpretable manner.

From an applicant's perspective, the findings indicate that employer attractiveness is shaped by a combination of tangible and intangible factors, including organisational culture, clarity around career development and alignment with personal values. These drivers vary meaningfully across respondent segments, highlighting the limitations of one-size-fits-all interpretations and underscoring the importance of segmentation-aware analysis in understanding applicant preferences.

From a surveyor's perspective, the results illustrate that survey design and quality directly influence respondent engagement and data completeness. By diagnosing issues such as question redundancy, cognitive burden, and structural drop-off early in the survey lifecycle, research teams can design more efficient instruments that preserve analytical depth while improving completion rates.

Taken together, the findings offer clear recommendations for practice. For employers, improving attractiveness perceptions may require reducing informational ambiguity by communicating clearer role expectations, career pathways and hiring processes, rather than focusing narrowly on compensation. For survey teams, prioritising key analytical questions earlier in the survey, minimising repetitive or low-value items, and optimising mobile usability can meaningfully reduce dropout and improve data quality.

Overall, the project demonstrates how analytics can shift survey practice from descriptive reporting toward insight-driven decision support. By combining data analytics with human-centred design principles, the proposed solution transforms raw survey data into meaningful intelligence for internal use. Future extensions could include real-time monitoring dashboards, deeper integration of large language models for automated reporting, and adaptive survey designs that dynamically respond to respondent behaviour.